DSC-002: Mathematical Methods

Lecture 4: **Inner Product Spaces** (Linear Algebra)

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Complex Numbers - Some Properties, Remarks

- **(1).** If $a + ib$ and $c + id$ are 2 complex numbers, then their -

  sum is : $(a + c) + i(b + d)$,

  product is : $(ac - bd) + i(ad + bc)$ (since $i^2 = -1$),

  quotient is : $\frac{(a+ib)}{(c+id)} = \frac{a+ib}{c+id} \cdot \frac{c-id}{c-id} = \frac{(ac+bd)+i(bc-ad)}{c^2+d^2}$.

Note: A complex number $(a + ib)$ can be viewed as the ordered pair (a, b) on the (complex) plane, so $(a^2 + b^2)^{1/2}$ is its distance from the origin.

NB: For more details on Complex Number, see "**Handout_1_Complex Number.pdf**", posted in LA Readings section in Piazza

Complex Numbers - Conjugate

◦ **(2).** The **Conjugate** of a complex scalar $(a + ib)$ is defined as $(a - ib)$, denoted by $\overline{a + ib}$, and their sum and product are real numbers.

◦ **(3). Conjugate Transpose** of a vector:
Let $y = (i, 2 - i, 3)^T$ be a (column) vector in C^3.
Take its transpose, then take the conjugate. $\overline{y^T} = (-i, 2 + i, 3)$. (row vector)

- We get the same result if we first take the conjugate of the vector and then take its transpose.
- The 'conjugate transpose' of a vector y is denoted $\mathbf{y}^*$.
  For real spaces, y^* is just y^T.

Inner Product

Let V be a real or complex vector space over a field F of scalars (think $\Re^n$ over $\Re$ or C^n over C; but spaces of real-valued or complex valued functions are other important examples)

Definition

A function form $f : V \times V \to F$ is an **inner product** on V if -

[i.e. for every ordered pair (x, y) of vectors, we denote $f(x, y)$ by $\langle x, y \rangle$]

(i) Conjugate symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$

(ii) Linearity: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle, \ \forall \alpha, \beta \in F, \ \forall x, y, z \in V$

(iii) Nonnegativity: $\langle x, x \rangle \geq 0$, and Nondegeneracy: with '=' *iff* $x = 0$.

Example

Let V be a real or complex n-dimensional v.s. Then the (scalar-valued) function $\langle x, y \rangle$ is an **inner product** *on V, and defined by –*

$\langle x, y \rangle = y^* x = \sum_{k=1}^{n} x_k \bar{y}_k$, *for every (ordered) pair of vectors x, y*

Remarks

Conjugate symmetry - Proof:

$$\overline{\langle y, x\rangle} = \overline{\sum_1^n y_k \bar{x_k}} = \sum_1^n \overline{y_k \bar{x_k}} = \sum_1^n \bar{y_k} x_k = \langle x, y\rangle$$

Note that if V is a real space, the definition of inner product reduces to the familiar **dot product**.

Remark (1). Since there's no difference between a **real number** and its conjugate, if V is a real space then $\overline{\langle y, x\rangle} = \langle y, x\rangle$, therefore condition (i) of "Conjugate Symmetry" in Inner Product definition is just *symmetry*.

Remark (2). Notice that $\langle x, y\rangle = y^* x$,
where x is taken as a column vector, and y^* is a row vector (transpose of a column vector).

Remarks

Remark (3). The benefit of conjugation is that for all $x \in V$, where $\langle x, x \rangle$ is a real number. Indeed,

$$\begin{aligned}\langle x, x \rangle = x^* x &= (a_1 - ib_1, \ldots, a_n - ib_n)(a_1 + ib_1, \ldots, a_n + ib_n)^T \\ &= \sum_{k=1}^{n} (a_k^2 + b_k^2)\end{aligned}$$

This helps us to define, for each $x \in V$, its **Norm** (or *distance from the origin*) as a real number.

Definition - Euclidean Norm, Orthogonality

Let V be a real or complex vector space.

Definition (Euclidean Norm)

The Euclidean Norm of a vector $x \in V$, denoted by $||x||$, is the real number $\langle x, x \rangle^{1/2}$. [Positive sq root]

We say that our inner product **induces** the above norm.
If $x = (x_1, ..., x_n)^T$, where $x_k = a_k + ib_k$,
we write $||x|| = (\sum_1^n |x_k|^2)^{1/2}$, where $|x_k|^2 = (a_k^2 + b_k^2)$.

For real spaces, $\langle x, x \rangle$ is the usual Euclidean distance $(\sum_1^n x_k^2)^{1/2}$.

Definition (Orthogonality)

2 vectors $u, v \in V$ are called **orthogonal** (denoted by $u \perp v$) if $\langle u, v \rangle = 0$ i.e. inner product is zero.

Remarks

Remark (4). Notation and Terminology. Treil uses (x, y) for inner product; since this is also notation for ordered pair, perhaps more preferable is $\langle x, y \rangle$. Ideally we should be consistent with textbook, BUT...

Remark (5). A vector space V along with some inner product defined on it is an **inner product space** or a **pre-Hilbert space**.

Once the inner product induces a *norm*, we can work with notions of convergence of sequences etc.

A pre-Hilbert space that is "complete" (w.r.t. the induced norm) is called a **Hilbert space**. (A pre-Hilbert space is complete if every Cauchy sequence in it converges to a vector in the space itself).

Properties of Inner Product

1. $\langle x, \alpha y + \beta z \rangle = \bar{\alpha}\langle x, y \rangle + \bar{\beta}\langle x, z \rangle$, $\forall\ \alpha, \beta \in F$, and $x, y, z \in V$.
   Proof. $(\alpha y + \beta z)^* x = \bar{\alpha} y^* x + \bar{\beta} z^* x$.

2. Let $x \in V$. Then $\langle x, y \rangle = 0$, $\forall y \in V$ *iff* $x = 0$.
   Proof. If $x = 0$, then $\langle x, y \rangle \equiv y^* x = 0$. On the contrary, if $x \neq 0$, then there exists y, namely $y = x$, s.t. $\langle x, y \rangle = \langle x, x \rangle > 0$.
   $\longmapsto$Every vector is perpendicular to origin!!

3. Let $x, y \in V$. Then $\langle x, z \rangle = \langle y, z \rangle$, $\forall z \in V$ *iff* $x = y$.
   Proof. By (2) above, $\langle x - y, z \rangle = 0\ \forall z \in V$ *iff* $(x - y) = 0$ or $x = y$. But $\langle x - y, z \rangle = 0$, by linearity of the inner product, means $\langle x, z \rangle - \langle y, z \rangle = 0$.

4. Suppose $A, B : X \to Y$ are two LTs or matrices that satisfy $\langle Ax, y \rangle = \langle Bx, y \rangle$, for all $x \in X$ and all $y \in Y$. Then $A = B$.
   Proof. For fixed x, property (3) implies $Ax = Bx$. Since we get this no matter what x we fix, the mappings A, B are *identical*.

Properties of Inner Product ... contd...

Theorem (Property 7: Cauchy-Schwarz Inequality)

$|\langle x, y\rangle| \leq ||x||\ ||y||, \forall x, y \in V.$

Proof

- Let $x, y \in V$ and $t \in F$. Then statement is trivially true if x or $y = 0$.
- Now, $0 \leq ||x - ty||^2 = \langle x - ty, x - ty\rangle = \langle x, x - ty\rangle - \langle ty, x - ty\rangle$
$= \langle x, x\rangle - \bar{t}\langle x, y\rangle - t\langle y, x\rangle + \bar{t}\langle ty, y\rangle$
$= ||x||^2 - \bar{t}\langle x, y\rangle - t\langle y, x\rangle + |t|^2\ ||y||^2$

Now the above expression is a quadratic function in t. We can minimize the expression w.r.to t in two ways –
(a) for a real-valued inner product space, and
(b) for complex-valued case.
[Proof continues...]

□

See next two slides...

Cauchy-Schwartz Inequality- Proof. (continues)

- **Minimization: The Real Case (Using Calculus)**

In the case of a **real** inner product space, the expression simplifies. Since $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ and t is real, we get:

$$Q(t) = \|\mathbf{x}\|^2 - 2t\langle \mathbf{x}, \mathbf{y} \rangle + t^2 \|\mathbf{y}\|^2 \geq 0$$

This is a quadratic function in t that opens upwards ($\|\mathbf{y}\|^2 > 0$). Its minimum occurs where its derivative is zero:

$$\frac{dQ}{dt} = -2\langle \mathbf{x}, \mathbf{y} \rangle + 2t\|\mathbf{y}\|^2$$

$$\text{Set } \frac{dQ}{dt} = 0: \quad -2\langle \mathbf{x}, \mathbf{y} \rangle + 2t_{min}\|\mathbf{y}\|^2 = 0$$

$$t_{min} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2}$$

◦ Minimization: The General Complex Case (Optimal Choice)

For a **complex** inner product space, we consider the inequality:

$$0 \leq \|\mathbf{x}\|^2 - t\langle \mathbf{y}, \mathbf{x}\rangle - \overline{t}\langle \mathbf{x}, \mathbf{y}\rangle + |t|^2\|\mathbf{y}\|^2,$$

which must hold for all $t \in \mathbb{C}$. To find the optimal t that minimizes the right-hand side, we complete the square.
Rewrite the expression:

$$|t|^2\|\mathbf{y}\|^2 - t\langle \mathbf{y}, \mathbf{x}\rangle - \overline{t}\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{x}\|^2.$$

Factor out $\|\mathbf{y}\|^2$ (assuming $\mathbf{y} \neq \mathbf{0}$):

$$\|\mathbf{y}\|^2 \left(|t|^2 - t \cdot \frac{\langle \mathbf{y}, \mathbf{x}\rangle}{\|\mathbf{y}\|^2} - \overline{t} \cdot \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2} \right) + \|\mathbf{x}\|^2.$$

Now, complete the square in t. Note that for complex t, we have:

$$\left| t - \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2} \right|^2 = |t|^2 - t \cdot \frac{\langle \mathbf{y}, \mathbf{x}\rangle}{\|\mathbf{y}\|^2} - \overline{t} \cdot \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2} + \frac{|\langle \mathbf{x}, \mathbf{y}\rangle|^2}{\|\mathbf{y}\|^4}.$$

[Continuing from last slide]
Therefore, the expression becomes:

$$\|\mathbf{y}\|^2 \left(\left| t - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \right|^2 - \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{y}\|^4} \right) + \|\mathbf{x}\|^2.$$

Simplifying:

$$\|\mathbf{y}\|^2 \left| t - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \right|^2 - \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2} + \|\mathbf{x}\|^2.$$

So,

$$0 \leq \|\mathbf{y}\|^2 \left| t - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \right|^2 + \left(\|\mathbf{x}\|^2 - \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2} \right).$$

The minimum occurs when the first term is zero, i.e., when

$$t = t_{\text{opt}} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2}.$$

Substituting this value yields the Cauchy-Schwarz inequality.

Minimization: The General Complex Case

Optimal Choice via Differentiation

- **Minimization via Wirtinger Derivatives**

For a **complex** inner product space, we minimize the function:

$$f(t) = \|\mathbf{x} - t\mathbf{y}\|^2 = \|\mathbf{x}\|^2 - t\langle \mathbf{y}, \mathbf{x}\rangle - \overline{t}\langle \mathbf{x}, \mathbf{y}\rangle + |t|^2\|\mathbf{y}\|^2.$$

To find the minimizer, we use the **Wirtinger derivatives**:

$$\frac{\partial f}{\partial t} \quad \text{and} \quad \frac{\partial f}{\partial \overline{t}}.$$

Treat t and $\overline{t}$ as independent variables. Then:

$$\frac{\partial f}{\partial \overline{t}} = -\langle \mathbf{x}, \mathbf{y}\rangle + t\|\mathbf{y}\|^2.$$

Set $\frac{\partial f}{\partial \overline{t}} = 0$ to find the critical point:

$$-\langle \mathbf{x}, \mathbf{y}\rangle + t\|\mathbf{y}\|^2 = 0 \quad \Rightarrow \quad t_{\text{opt}} = \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2}.$$

Cauchy-Schwartz Inequality- Proof. (final)

Substitute $t = \frac{\langle \mathbf{x},\mathbf{y} \rangle}{\|\mathbf{y}\|^2}$ into the inequality:

$$
\begin{aligned}
0 &\leq \|\mathbf{x}\|^2 - \left(\frac{\langle \mathbf{x},\mathbf{y} \rangle}{\|\mathbf{y}\|^2}\right)\langle \mathbf{y},\mathbf{x} \rangle - \overline{\left(\frac{\langle \mathbf{x},\mathbf{y} \rangle}{\|\mathbf{y}\|^2}\right)}\langle \mathbf{x},\mathbf{y} \rangle + \left|\frac{\langle \mathbf{x},\mathbf{y} \rangle}{\|\mathbf{y}\|^2}\right|^2 \|\mathbf{y}\|^2 \\
&= \|\mathbf{x}\|^2 - \frac{\langle \mathbf{x},\mathbf{y} \rangle\langle \mathbf{y},\mathbf{x} \rangle}{\|\mathbf{y}\|^2} - \frac{\langle \mathbf{y},\mathbf{x} \rangle\langle \mathbf{x},\mathbf{y} \rangle}{\|\mathbf{y}\|^2} + \frac{|\langle \mathbf{x},\mathbf{y} \rangle|^2}{\|\mathbf{y}\|^4}\|\mathbf{y}\|^2 \\
&= \|\mathbf{x}\|^2 - \frac{|\langle \mathbf{x},\mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2} - \frac{|\langle \mathbf{x},\mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2} + \frac{|\langle \mathbf{x},\mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2} = \|\mathbf{x}\|^2 - \frac{|\langle \mathbf{x},\mathbf{y} \rangle|^2}{\|\mathbf{y}\|^2}
\end{aligned}
$$

Rearranging this inequality immediately gives us the desired result:

$$
\begin{aligned}
|\langle \mathbf{x},\mathbf{y} \rangle|^2 &\leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2 \\
\implies |\langle \mathbf{x},\mathbf{y} \rangle| &\leq \|\mathbf{x}\| \|\mathbf{y}\| \qquad \blacksquare
\end{aligned}
$$

This completes the proof. The choice $t = \frac{\langle \mathbf{x},\mathbf{y} \rangle}{\|\mathbf{y}\|^2}$ is justified as it provides the strongest possible inequality from $\|\mathbf{x} - t\mathbf{y}\|^2 \geq 0$.



Properties of Inner Product ... contd...

Theorem (Properties of Triangle Inequality)

$$||x+y|| \leq ||x|| + ||y||, \ \forall x, y \in V$$

Proof

$$||x+y||^2 = \langle x+y, x+y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle$$

By Cauchy's inequality, both $\langle x, y \rangle$ and $\langle y, x \rangle$ must be $\leq ||x||\ ||y||$.

So the RHS is $\leq ||x||^2 + 2||x||\ ||y|| + ||y||^2$. □

We can define a norm more generally than the Euclidean norm.

Definition - Norm

A **norm** on a vector space V is a function that associates, with each $v \in V$, a real number $||v||$ s.t. –

(i) Homogeneity: $||\alpha v|| = |\alpha| \cdot ||v||, \ \forall v \in V, \alpha \in F$

(ii) Triangle Inequality: $||u + v|| \leq ||u|| + ||v||, \ \forall u, v \in V$

(iii) Nonnegativity: $||v|| \geq 0$, $\forall v$ and Non-degeneracy: with ' $=$' *iff* $v = 0$

Example: Prove that $u \perp v \leftrightarrow v \perp u$.

To start, let's assume $u \perp v$, i.e. $\langle u, v \rangle 0 \implies = v^* u = 0$.

Taking conjugate transpose of both sides of the above expression,

$$(v^* u)^* = u^* (v^*)^* = u^* v = \overline{v^* u} = \overline{0} = 0,$$

i.e. $\langle v, u \rangle = 0$ i.e. $v \perp u$.

So $u \perp v$ *iff* $v \perp u$.

A vector v is orthogonal to a subspace E if $v \perp w,\ \forall w \in E$.

i.e. v is perpendicular to every vector in the subspace E.

Let $span(\{w_1, ..., w_r\}) = E$. Then $v \perp E$ iff $v \perp w_k,\ \forall k = 1, ..., r$.

i.e. v is perpendicular to every "basis" vector in the subspace E.

Suppose $v \perp w_k,\ \forall k = 1, ..., r$ and let any $w \in E$.
So, $w = \sum_1^r c_k w_k$ for some c_k's.
So, $\langle w, v \rangle = \langle \sum_1^r c_k w_k, v \rangle = \sum_1^r c_k \langle w_k, v \rangle = 0$. Thus, $v \perp w, \forall w \in E$.
Conversely, v not $\perp w_i$, for some i, implies v is not orthogonal to E. □

Ex: Consider a vector $v = (0, 0, z)^T \in \Re^3$ which is $\perp$ to X-Y plane.
Now verify the above statement.

Orthogonal Set of Vectors: A set of vectors $\{v_1, ..., v_n\}$ is orthogonal if every pair $[v_i, v_j]$, for $i \neq j$ we have: $v_i \perp v_j, \quad \forall i, j \in \{1, ..., n\}, \quad i \neq j$.

A set of vectors $\{v_1, ..., v_n\}$ is non-zero & orthogonal implies that $\{v_1, ..., v_n\}$ linearly independent.

Simply let $\sum c_i v_i = 0$. (i.e. an LC of some vector $= 0$)

So, $\langle \sum c_i v_i, v_j \rangle = 0$ for any v_j. (Why?)

Now, $\langle \sum c_i v_i, v_j \rangle = \sum_{i \neq j} c_i \langle v_i, v_j \rangle + c_j ||v_j||^2$,

$= 0 + c_j ||v_j||^2$ (due to orthogonality v_i, v_j)

Since $||v_j||^2 > 0$, this implies $c_j = 0$. Thus, all c's are 0.

Since this is true irrespective of choice of j, LI follows. □

The Representation (in Orthogonal Basis)

Suppose we represent a vector $v \in V$ in terms of a basis $\{v_1, ..., v_n\}$, as $v = \sum c_i v_i$. To find the **Representation** (i.e. the c_i's) we have to solve a system of linear equations, namely $Ac = v$, where $A = (v_1 \ v_2 \ ... \ v_n)$ is the matrix whose columns are the basis vectors, and $c = (c_1, ..., c_n)^T$.

But, if the basis is orthogonal, and $v = \sum c_i v_i$, we don't need to solve a linear system to find the c_i's. Instead, notice that for any $k = 1, ..., n$,

$$\langle v, v_k \rangle = \langle \Sigma c_i v_i, v_k \rangle = \sum_{i \neq k} c_i \underbrace{\langle v_i, v_k \rangle}_{=0} + c_k ||v_k||^2,$$

which, due to orthogonality, equals to $c_k ||v_k||^2$.

So, we have: $c_k = \frac{\langle v, v_k \rangle}{||v_k||^2}, \ \forall k = 1, ..., n.$

Ex: Consider an orthogonal basis $\{(1,1)^T, (-1,1)^T\}$ in V^2.

Now for a vector $v = (6,5)^T$ find the co-ordinate represention in V following the mothod above.

Definition – Orthogonal Projection

Let $v \in V$ and E a subspace of V. The **orthogonal projection** of v onto E, denoted $P_E v$, is the vector $w \in E$ s.t. $(v - w) \perp E$.

This presumes an orthogonal projection always exists and is unique. Visually, we are dropping a perpendicular from v to a plane or hyperplane E. If $P_E v$ exists, it is unique.

$P_E v$ minimises distance from v to E:
$||v - P_E v|| \leq ||v - x||, \ \forall x \in E,$
and '$=$' $\Rightarrow x = P_E v$.

Proof. [Draw picture.] *Let $P_E v \equiv w$. Since E is a subspace, and $w, x \in E$, so $(w - x) \in E$. Thus $(v - w) \perp E$, and $(v - w) \perp (w - x)$.*

Applying Pythagorean theorem to the right triangle Δvwx,

$$||v - x||^2 = ||v - w||^2 + ||w - x||^2 \geq ||v - w||^2.$$

Thus the equality holds iff $x = w$.

Deriving Orthogonal Projection

Case 1: Projection onto a subspace spanned by ONE vector

◦ Draw the picture in $\Re^2$.

Let v be a vector and $E = span\{a\}$ be a subspace in $\Re^2$.

Projection of v onto E happening at w point i.e. $P_E v = w$.

This w vector is α proportion of vector a in E, i.e. $w = a \cdot \alpha \cdots (1)$

By construction, $(v - w) \perp E \Rightarrow (v - w) \perp a$.

$\Rightarrow \langle v - w, a \rangle = 0 \Rightarrow \langle v, a \rangle - \langle w, a \rangle = 0$

$\Rightarrow a^* v - a^* w = 0$

$\Rightarrow a^* v - a^* . \alpha a = 0 \quad$ [replacing $w = \alpha a$]

$\therefore \alpha = \frac{a^*}{a^* a} v$

Putting the value of α in Eq. (1), we get –

$w = a \frac{a^*}{a^* a} v = \frac{a a^*}{a^* a} v.$

So, $P_E = \frac{a a^*}{a^* a}$

Note: Numerator is a matrix, and denominator is scalar.

Deriving Orthogonal Projection

Case 2: Projection onto a subspace spanned by MANY vectors

◦ Draw the picture in $\Re^3$.

Now, suppose we are in $\Re^3$. A vector v is orthogonally projected on X-Y plane i.e. a subspace E spanned by two vectors $\{a, b\} := \mathscr{A}$.

Note: the vectors $\{a, b\}$ may or may not be orthogonal basis in $\mathscr{A}$.

- **if non-orthogonal but LI**, then general case possible
  - The projection $w = P_E v$ must satisfy:
    1. $w \in E$, so $w = \mathcal{A}\alpha$ for some $\alpha \in \mathbb{R}^k$ (or $\mathbb{C}^k$).
    2. The error $(v - w)$ is orthogonal to E, i.e., $\mathcal{A}^*(v - w) = 0$.
    3. Continuing the process as before, we have

$$P_E = \frac{\mathscr{A}\mathscr{A}^*}{\mathscr{A}^*\mathscr{A}} = \mathscr{A}\frac{1}{\mathscr{A}^*\mathscr{A}}\mathscr{A}^* = \mathscr{A}\left(\mathscr{A}^*\mathscr{A}\right)^{-1}\mathscr{A}^*$$

- **if orthogonal**, see computation of Projection matrix as in Note (1) in pp.26 and Note (2) in pp.27.

Examples

Example 1: In $\Re^2$, project $v = (4,5)^T$ on the surface spanned by $(1,2)^T$. Find the projection matrix and the projection point.
We derived earlier, $P_E = \frac{aa^*}{a^*a}$

Now, numerator $aa^* = \begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}1 & 2\end{pmatrix} = \begin{pmatrix}1\cdot 1 & 1\cdot 2\\ 2\cdot 1 & 2\cdot 2\end{pmatrix} = \begin{pmatrix}1 & 2\\ 2 & 4\end{pmatrix}$ notice the rank

Now, calculate P_E and $P_E v$.
TRY also, to project the vector onto X-axis and Y-axis.

Example 2: In $\Re^3$, Project $(1,1,1)^T$ onto a surface E, spanned by $(1,0,1)^T$ and $(0,1,1)^T$.

Special Case: Orthogonal Basis

If the columns of $\mathcal{A}$ are **orthonormal**, then $\mathcal{A}^*\mathcal{A} = I$, and:

$$P_E = \mathcal{A}\mathcal{A}^*.$$

Properties of Projection Matrix

Suppose that P is the matrix of orthogonal projection. Then –

1. $P^T = P$ i.e. projection matrix is symmetric
2. $P^2 = P$ i.e. P is idempotent
3. $\det(P) = 0$ i.e. rank of P is less than the number of columns

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Try to prove these properties
(not numerically, but using the expression of P).

Computing Orthogonal Projection (on a subspace with Orthogonal basis)

Proposition.
Let $\{w_1, ..., w_r\}$ *be an orthogonal basis for subspace* E, $w \in E$ and $v \in V$.
Then $P_E v = w = \sum_{k=1}^{r} \alpha_k w_k$, *where* $\alpha_k = \frac{\langle v, w_k \rangle}{||w_k||^2}$, $\forall k = 1, ..., r$.

Proof

- Draw the picture in $\Re^2$ or $\Re^3$.

Write $P_E v \equiv w$. Since $w \in E$, $w = \sum_{k=1}^{r} c_k w_k$, where $c_k = \frac{\langle w, w_k \rangle}{||w_k||^2}$.

Finally, since $v = w + (v - w)$ and $(v - w) \perp w_k$, $\forall k = 1, ..., r$, we have $\langle v, w_k \rangle = \langle w + v - w, w_k \rangle = \langle w, w_k \rangle + \langle v - w, w_k \rangle = \langle w, w_k \rangle$. □

Note.(1) Basically, the scalars in the LC for the projection point w (of v) involve $\langle v, w_k \rangle$ rather than $\langle w, w_k \rangle$ because $v = w + (v - w)$, thus w and $(v - w)$ has zero inner product with vectors in E, being orthogonal.

Computing Orthogonal Projection (on a subspace with Orthogonal basis)

Note.(2) $P_E = \sum_{k=1}^{r} \frac{1}{||w_k||^2} w_k w_k^*$ [where $\{w_1, ..., w_r\}$ is an ortho. basis in E]. If V is n-dimensional, w_k is a $n \times 1$ vector, so that $w_k w_k^*$ is an $n \times n$ matrix. Indeed,

$$P_E\ v = \sum \frac{\langle v, w_k \rangle}{||w_k||^2} w_k = \sum \frac{1}{||w_k||^2} w_k . \langle v, w_k \rangle = \left[\sum \frac{1}{||w_k||^2} w_k w_k^*\right]\ v.$$

The 2nd equality follows since $\langle v, w_k \rangle$ is a scalar.

Note.(3) The projection P_E is an LT ($P_E : V \to V$), note $E \in V$.

Indeed, for vectors v, v' and scalars c_1, c_2, we have

$$P_E(c_1 v + c_2 v') = \sum_{k=1}^{r} \frac{\langle c_1 v + c_2 v', w_k \rangle}{||w_k||^2} w_k .$$

By linearity of inner product, this

$$= c_1 \sum \frac{\langle v, w_k \rangle}{||w_k||^2} w_k + c_2 \sum \frac{\langle v', w_k \rangle}{||w_k||^2} w_k = c_1 P_E v + c_2 P_E v'.$$

Definition – Orthogonal Complement, Decomposition

Definition

Let V be a v.s. and E be a subspace of V. Then the **Orthogonal Complement** of E in V, denoted $E^{\perp}$ (pronounced, E *perp*) is the set of vectors **u** such that **u** is orthogonal to all vectors in E i.e. $\{u \in V | u \perp E\}$.

Note.(4) (i) $E^{\perp}$ is a subspace.
(ii) $(E^{\perp})^{\perp} = E$. [see Handout 2 for proof]

Note.(5) $v = P_E v + (v - P_E v)$ for any $v \in V$

Above is the unique representation of v as the sum of 2 vectors, one lying in E [namely $P_E v$] and the other [namely $(v - P_E v)$] lying in $E^{\perp}$. Uniqueness results from the uniqueness of $P_E v$.

Thereby, any v.s. V can be decomposed into two parts, $V = E \oplus E^{\perp}$.

Example: In $\Re^2$, $(x_1, x_2)^T = (x_1, 0)^T + (0, x_2)^T$ is the simplest such representation, with E being the horizontal axis.

The Annihilator Matrix and Residuals

Measuring What's Left Unexplained

Recall: Projection Matrix For a real matrix X with full column rank, the projection matrix onto $\text{col}(X)$ is: $P = X(X^TX)^{-1}X^T$
This maps any vector $\mathbf{y}$ to its projection $\hat{\mathbf{y}} = P\mathbf{y}$.

The **annihilator matrix** (or residual maker matrix) is defined as:

$$M = I - P = I - X(X^TX)^{-1}X^T$$

where I is the identity matrix.

- M projects onto the **orthogonal complement** of $\text{col}(X)$
- Applying M to $\mathbf{y}$ gives the **residual vector**:

$$\mathbf{e} = M\mathbf{y} = (I - P)\mathbf{y} = \mathbf{y} - \hat{\mathbf{y}}$$

- M is also **symmetric** ($M^T = M$) and **idempotent** ($M^2 = M$)

Orthogonal Complement - Example

Ex-1. Let $V = \Re^2$ and W be the subspace spanned by $(1, 2)$. Then $W^{\perp}$ is the set of vectors (a, b) with

$$(a, b) \cdot c(1, 2)^T = 0 \text{ , (where } \mathbf{c} \neq 0 \text{ be some constant)}$$

or, $ac + 2bc = 0 \Rightarrow a + 2b = 0.$

This is a 1-dimensional vector space spanned by $(-2, 1)$.

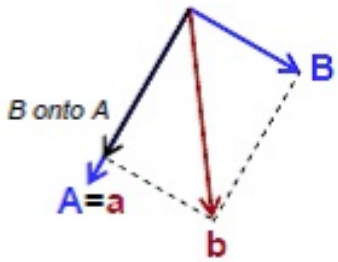
In the example above the orthogonal complement was a subspace. This will always be the case.

Ex-2. In V^3, suppose XY-plane is a subspace, what is the orthogonal complement of that?

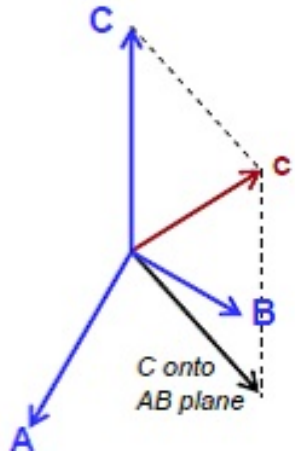
For more examples on this please refer **Handout 2 Inner Product Space (Numerical example)**.

Orthogonalisation: The Gram Schmidt Procedure

- **The Gram-Schmidt orthogonalization procedure operates as follows**
  - Start with A=a
    - This gives the first direction
  - The second direction must be perpendicular to A
    - Start with B=b and subtract its projection along A
    - This leaves the perpendicular part, which is the orthogonal vector B



  - The third direction must be perpendicular to A and B
    - Start with C=c and subtract its projections along A and B
  - The resulting vectors {A,B,C} are orthogonal and span the same space as {a,b,c}



Numerical Example of Gram-Schmidt procedure

Apply Gram-Schmidt procedure to orthogolize the vectors:
$x_1 = (1, 2, -2)^T$, $x_2 = (1, -1, 4)^T$ and $x_3 = (2, 1, 1)^T$.

- Let $X_1 = x_1$, thus $E_1 = span\{X_1\}$.
- Get $X_2 \perp E_1$ as follows –
  Define $X_2 = x_2 - P_{E_1} x_2$
  Since X_1 is the sole basis vector of E_1, then –
  $P_{E_1} x_2 = \alpha_1 . X_1$, where $\alpha_1 = \frac{\langle x_2, X_1 \rangle}{||X_1||^2}$.
  Calculate $\langle x_2, X_1 \rangle$ and $||X_1||^2$, then X_2.
- Let $E_2 = span\{X_1, X_2\}$, where $X_1 \perp X_2$
  - Define $X_3 = x_3 - P_{E_2} x_3 = x_3 - \frac{\langle x_3, X_1 \rangle}{||X_1||^2} . X_1 - \frac{\langle x_3, X_2 \rangle}{||X_2||^2} . X_2$
  - i.e. $P_{E_2} x_3 = \beta_1 . X_1 + \beta_2 . X_2$, where, $\beta_1 = \frac{\langle x_3, X_1 \rangle}{||X_1||^2}$ and $\beta_2 = \frac{\langle x_3, X_2 \rangle}{||X_2||^2}$
- Compute all these and find the orthogonalized vectors.

Application - Least Squares ... I

You have a model which explains the wage y of an individual as a function of k explanatory variables, such as years of schooling, gender etc.

Say, $y = \sum_j x_j \beta_j + \epsilon$, where $\epsilon, x_j's$ are random variables.
This is the *true model* in the population but you do not know the β_j's.

One way to estimate them is as follows:

You collect data on say the wages of n individuals, y_i equals wage of individual i, as well as on k variables that may explain wage $(x_{i1}, ..., x_{ik}) \equiv x_i^T$, for individual '$i$'. Typically, $n >> k$.

You want β_i's that minimise $\sum_{i=1}^{n} (y_i - x_i^T \hat{\beta})^2$,
i.e. minimizing the sum of squares of errors. (**But why so?**)

Least Squares ... II

That means, choose $\hat{\beta} = (\hat{\beta_1}, ..., \hat{\beta_k})^T$ to minimise $||y - X\hat{\beta}||^2$, where $y = (y_1, ..., y_n)^T$, and X is the $n \times k$ matrix whose j^{th} column contains observations x_{ij} for all individuals i.

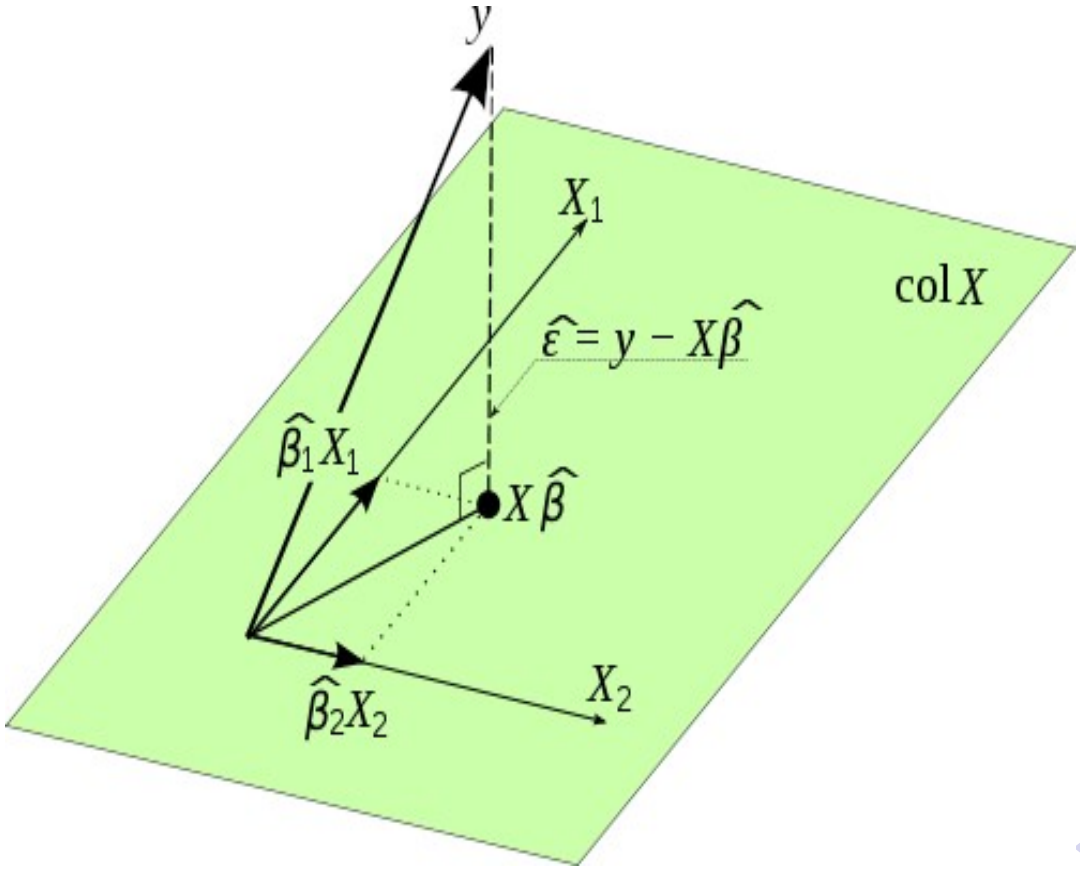
Since $||y - X\hat{\beta}||^2 \geq 0$, this is minimised *iff* $||y - X\hat{\beta}||$ is minimised.

If we can find $\hat{\beta}$ that solves the linear system $y = X\hat{\beta}$ we are done.
But since $n >> k$, $Rank(X) \leq k < n$, and $y \in \Re^n$,
it is '**unlikely**' that $y \in R(X)$, the span of the columns of X.
(Note $X : \Re^k \to \Re^n$). [like in the **Example *** below.]

Example *: Find a solution when $X = \begin{pmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}$ and $Y = \begin{pmatrix} 2 \\ 0 \\ 11 \end{pmatrix}$.

Least Squares ... III

So the general problem is to find $\hat{\beta}$ s.t. $X\hat{\beta} = P_{R(X)}y = \hat{\mathbf{y}}$,
i.e. s.t. $X\hat{\beta}$ is the orthogonal projection of y onto $R(X)$, as this minimizes the distance between y and $R(X)$. (Notice that for any $\hat{\beta}$, $X\hat{\beta} \in R(X)$; we want the $X\hat{\beta}$ which is the closest point to y, from $R(X)$).



Least Squares ... IV

If we know an orthogonal basis, (or construct one using say Gram-Schmidt method) in the subspace $R(X)$, it is easy to get $P_{R(X)}y$; then just solve the linear system $X\hat{\beta} = P_{R(X)}y$ for $\hat{\beta}$. But constructing an orthogonal basis takes more calculations than the following simpler method.

If $X\hat{\beta}$ is the orthogonal projection of y onto $R(X)$, then

$$(y - X\hat{\beta}) \perp x_j, \text{ for all columns } x_j,\ j = 1, ..., k \text{ of } X,$$

because $y - X\hat{\beta}$ is orthogonal to every vector in $R(X)$.

So, $$x_j^*(y - X\hat{\beta}) = 0,\ \forall j = 1, ..., k.$$

Stacking the rows vertically for x_j^*, we have –

$$\Rightarrow \quad X^*_{k\times n}(y - X\hat{\beta}) = 0,$$

$$\text{or} \quad X^*X\hat{\beta} = X^*y.$$

This is a linear system amd called

the Normal Equations and can be solved for $\hat{\beta}$.

Least Squares ... V

If $(X^*X)_{k\times k}$ has full rank k (which happens *iff* X has full rank k), we get: $\hat{\beta} = (X^*X)^{-1}X^*y$. And, projn point: $X\hat{\beta} = X(X^*X)^{-1}X^*y$

Note that in econometrics, the "data matrix" X is likely to have only real entries, so that the "star" or conjugate transpose above can be replaced with a "T" or transpose.

Before we close the section, we note that the square matrix X^*X (which is $k \times k$) is invertible iff $\text{Rank}(X^*X) = k$. Furthermore, $\text{Rank}(X^*X) = \text{Rank}(X)$. Hence, X^*X is invertible iff the original data matrix X has full column rank k.

For, to be **invertible**, X^*X needs to be a bijection from $C^k \to C^k$.
If $Rank(X^*X) = k \Rightarrow R(X^*X) = C^k$, so X^*X is surjective.
Moreover, if 2 vectors $b_1 \neq b_2$, then $(X^*X)b_1 \neq (X^*X)b_2$.
Because, if $(X^*X)b_1 = (X^*X)b_2 \quad \Rightarrow \quad (b_1 - b_2) \in N(X^*X)$.
But $Rank(X^*X) = k \Rightarrow N(X^*X) = \{0\}$, so this can't be true.
So, X^*X is injective also. Therefore, X^*X is bijective.

We end by showing that: X^*X has full rank k *iff* X does.

Least Squares ... VI

Let X be $n \times k$, $n >> k$. Then X^*X is invertible iff $Rank(X) = k$.

We prove that: X^*X is invertible *iff* $Nullity(X^*X) = 0$.

We know: $Rank(X) = k$ *iff* $Nullity(X) = 0$.

So we are done if we show $Nullity(X^*X) = 0$ *iff* $Nullity(X) = 0$.

We show the stronger result that $N(X^*X) = N(X)$.

Let $b \in N(X)$, so $Xb = 0$. $\Rightarrow$ $X^*Xb = 0$, so $b \in N(X^*X)$.

Conversely, let $b \in N(X^*X)$. So, $X^*Xb = 0$.

So, $\langle X^*Xb, b\rangle = 0$ or $b^*X^*Xb = 0$ or $\langle Xb, Xb\rangle = 0$.

So, $Xb = 0$ or $b \in N(X)$. □

Identity Property of Adjoint Matrix

Definition

$$\langle Ax, y\rangle = \langle x, A^*y\rangle \quad \forall x, y \in \mathbb{C}^n$$

Proof

Now, to prove the main identity:

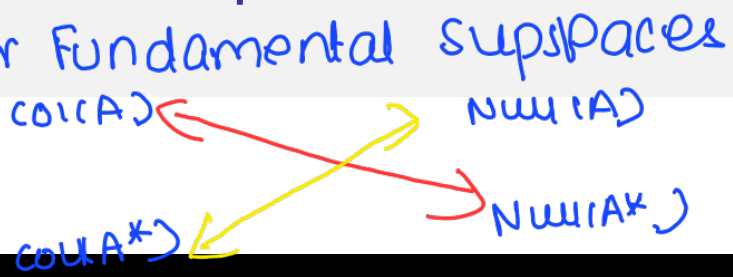
$$\langle Ax, y\rangle = \underbrace{y^*A}\, x = (A^*y)^*x = \langle x, A^*y\rangle,$$

the first and the last equalities here follow from the definition of inner product in F^n, and the middle one follows from the fact that $(A^*y)^* = y^*(A^*)^* = y^*A$. [Used here, $(AB)^* = B^*A^*$.]

Fact: Let us note that the adjoint operator is unique: if a matrix B satisfies $\langle Ax, y\rangle = \langle x, By\rangle$ $\forall x, y$ then $B = A^*$.
Indeed, by the definition of A^* we get $\langle x, A^*y\rangle = \langle x, By\rangle \quad \forall x$ and therefore by [Corollary 1.5 Treil p.118]: $A^*y = By$.
Since it is true for all y, the linear transformations, and therefore the matrices A^* and B coincide. □

Relation between Range and Null Spaces

Of A and A^*



Consider an LT $A : V \to W$

Theorem

$Ran(A^*) = (NullA)^{\perp}$.

Proof

$Null(A) = \{z | Az = 0\}$. $Az = 0$ *iff* $z^*A^* = 0^*$ (taking conjug transp).
That is, *iff* $z^*a_i = 0^*$, or $a_i^*z = 0$, $\forall$ columns a_i of A^*.
That is, *iff* $z \perp a_i$ for all columns a_i of A^*, *iff* $z \perp RanA^*$.
So, $Null(A) = \{z | z \perp RanA^*\}$. (In the real matrix case, just think of $N(A)$ as all vectors z orthogonal to the rows of A and hence to the columns of A^T and hence to $Ran(A^T)$). So, $Null(A) = (RanA^*)^{\perp}$.
Taking orthogonal complements on both sides, $(NullA)^{\perp} = RanA^*$. □

As a corollary, interchanging A and A^*, we get $(NullA^*)^{\perp} = RanA$.

Relation between Range and Null of A contd...

Consider an LT $A : V \to W$

Th

$Null(A^*) = (RanA)^{\perp}$.

Proof

Let $w \in W$. Then
$w \in null(A^*) \Leftrightarrow A^*w = 0$
Now, $\langle v, A^*w \rangle = 0$ for all $v \in V$ which is orth. comp. to $Null(A^*)$.
$\Leftrightarrow \langle Av, w \rangle = 0$ for all $v \in V$ (by Indentity property of Adjoint)
$\Leftrightarrow w \in (RanA)^{\perp}$.
Thus $Null(A^*) = (RanA)^{\perp}$ □

Note: If we take the orthogonal complement of both sides of the previous result, we then get $RanA = (Null(A^*))^{\perp}$.

Another take: $Null(A^*) = (RanA)^\perp$

We can show that the null space of the adjoint of A (denoted as A^*) equals the orthogonal complement of the range of A (denoted as $(\text{Ran}A)^\perp$).
First, for any vector $x \in null(A^*)$, i.e., $A^*x = 0$.

Next, consider the definition of the adjoint operator: $\langle A^*x, y\rangle = \langle x, Ay\rangle$

For x in $\text{Null}(A^*)$, this means: $\langle 0, y\rangle = \langle x, Ay\rangle = 0$

Thus, for all y, $0 = \langle x, Ay\rangle$ indicating that x is orthogonal to every vector in the range of A.
Hence, $x \in (\text{Ran}A)^\perp$.

Conversely, if $x \in (\text{Ran}A)^\perp$, then for any y, $\langle x, Ay\rangle = 0$
which means: $\langle A^*x, y\rangle = 0$
Therefore, $A^*x = 0$ and $x \in \text{Null}(A^*)$.

This establishes the equality: $\text{Null}(A^*) = (\text{Ran}A)^\perp$